

Pilatus Integration GUI

User Manual

SSRL Beamline 2-1 · Pilatus 100K · updated October 2026

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1. Overview

The Pilatus Integration GUI turns the Pilatus 100K images collected at SSRL beamline 2-1 into one-dimensional powder diffraction patterns (intensity vs 2θ) and displays them. The detector is scanned in 2θ during each measurement; the program combines every image of a scan into one pattern, normalized to the incident beam monitor, and writes it as an .xye file.

With it you can:

- integrate a single scan or a range of scans with one click;
- display patterns one at a time, overlaid, or as a contour (waterfall) plot;
- import patterns integrated earlier;
- run **live mode** during an experiment: each scan is integrated automatically as soon as it is complete and added to a live waterfall plot, with no clicks needed;
- **detect events**: flag scans where the pattern changed (a reaction, a phase transition, peaks sharpening or broadening), during a live run or afterwards on loaded data.

2. Before you start

You need four things. The first three come from the beamline; the fourth is wherever you want the integrated data to go.

Item	What it is
Calibration file (.cal)	Direct-beam position on the detector and the sample-to-detector distance. Made with the beamline calibration script from a direct-beam scan; beamline staff will tell you which file applies to your measurements. See Section 10.
SPEC file	The data file written by SPEC during your measurements (no file extension). It holds the 2θ position and monitor counts for every image, and your user name.
Image folder	The folder holding the Pilatus .raw images (and their .pdi files) for the scans in the SPEC file.
Output folder	Where the integrated .xye files are written. Defaults to the folder of the SPEC file.

Starting the program

On the beamline computer, double-click Pilatus_Integration_GUI.exe (or its shortcut). Keep manual.pdf in the same folder as the program so that **Help > Open Manual PDF** can find it.

To run from the Python source instead, open a terminal and run:

```
conda activate PilatusGUI
python Pilatus_Integration_GUI.py (in the program folder)
```

3. Program layout

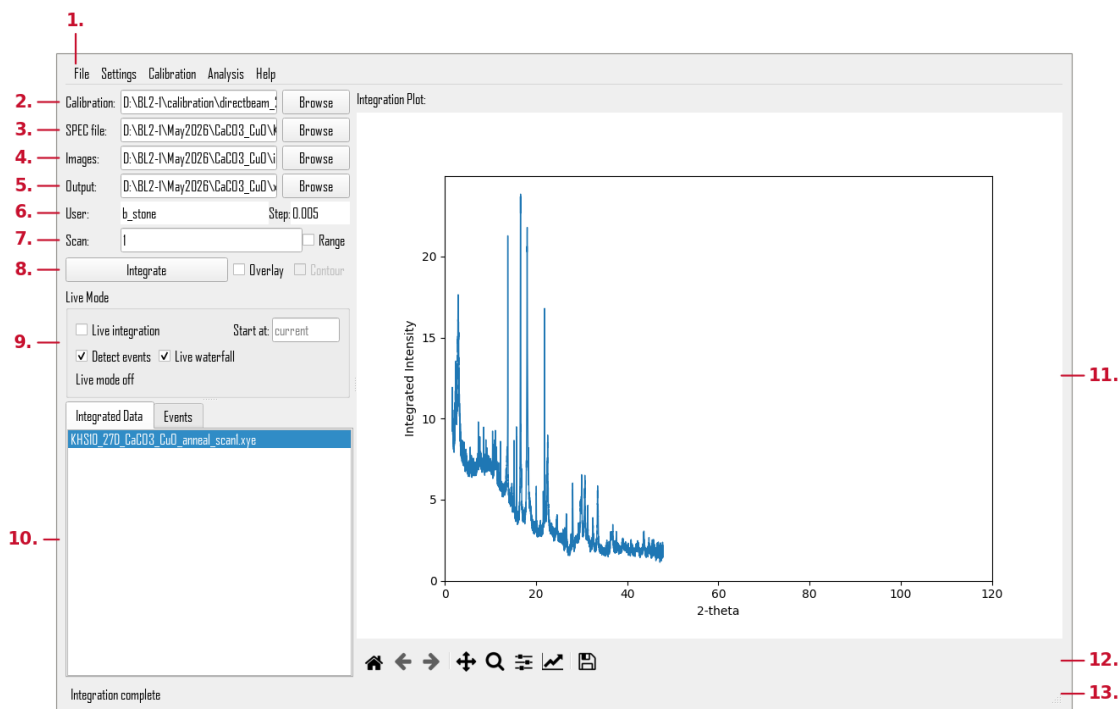


Figure 1. The main window after integrating one scan.

1.	Menu bar: File, Settings, Calibration, Analysis, and Help menus (Section 12).
2.	Calibration: the calibration file (.cal) to use for integration.
3.	SPEC file: the SPEC data file (no extension). Selecting it also fills in the user name and sets the output folder to the SPEC file's folder.
4.	Images: the folder holding the Pilatus images.
5.	Output: the folder where integrated .xye files are saved.
6.	User (read from the SPEC file) and Step (2 θ bin width, set in Settings > Integration Settings). Both are for information and cannot be edited here.
7.	Scan: the scan number to integrate. Tick Range to enter a first and last scan instead, for in-situ or operando series.
8.	Integrate starts integration of the scan or range. Overlay shows several patterns at once; Contour (with Overlay on and more than 4 patterns selected) shows them as a contour plot.
9.	Live Mode: automatic integration of each new scan during an experiment, with optional event detection (Section 8).
10.	Integrated Data tab: every pattern integrated or imported this session. Click a pattern to plot it. Events tab: detected events (Section 9).
11.	Plot area.
12.	Plot toolbar: reset view, back/forward, pan, zoom, subplot spacing, axis and line options, and save the figure as an image.
13.	Status bar: progress and messages.

The dividers between the left panel and the plot, and between the controls and the data list, can be dragged to resize the panels. If the window is small, the controls scroll.

4. Integrating data

4.1 Integrate one scan

1. Click **Browse** next to **Calibration** and choose the .cal file.
2. Click **Browse** next to **SPEC file** and choose your SPEC file. The **User** field fills in and **Output** is set to the same folder; change Output if you want the data elsewhere.
3. Click **Browse** next to **Images** and choose the folder with the Pilatus images.
4. Type the scan number in **Scan** and click **Integrate**.

A progress bar appears at the bottom of the window. When integration finishes, the pattern is saved to the output folder, added to the **Integrated Data** list, and plotted. A typical scan (16 images) takes well under a second.

4.2 Integrate a range of scans

Tick **Range**, enter the first and last scan numbers, and click **Integrate**. The scans are integrated one after another, in order, and each is saved and added to the list as it finishes. For scans that are still being collected, use live mode instead (Section 8).

4.3 Output files

Each scan is written as <SPEC file name>_scan<N>.xye in the output folder, for example KHS10_27D_CaCO3_CuO_anneal_scan1.xye. The file has three columns with no header:

Column	Contents
1	2 θ (degrees), bin centres at the chosen step size
2	Intensity: average pixel intensity in each 2 θ bin, normalized to the monitor (I0) for each image and scaled to the monitor counts of the first image (a rough counts-per-pixel scale)
3	Uncertainty σ , from the chosen error model (Section 7)

Note. Existing files with the same name are overwritten when a scan is integrated again.

5. Viewing data

Patterns in the **Integrated Data** list can be plotted in three ways:

- **Single:** click a pattern in the list to plot it on its own.
- **Overlay:** tick **Overlay**, then click patterns to add them to (or remove them from) the plot. Each pattern gets its own colour and a legend entry.

- **Contour:** with **Overlay** ticked and more than 4 patterns selected, tick **Contour** to show intensity as a function of 2θ and pattern number.

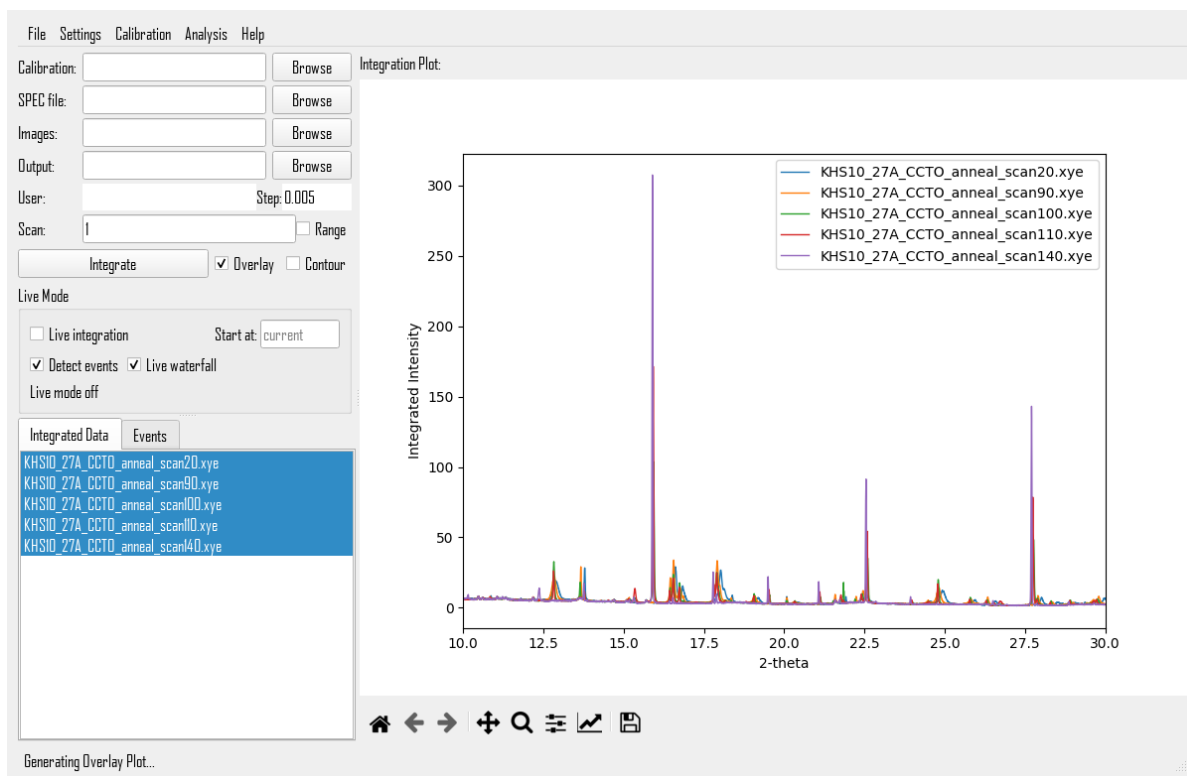


Figure 2. Overlay of five patterns from an in-situ run, zoomed to 10–30° with Settings > Plot Settings.

Plot settings

Settings > Plot Settings changes line width, style and colour, the colour map used for contour and waterfall plots, the marker style, the 2θ range (Min X, Max X; **Reset X-Axis** returns to 0–120°), and the intensity scale (linear, square root, or log). Click **Accept** to apply.

Line Width: 1.00

Line Style: solid

Line Color: Select Color

Colormap: viridis

Marker Style: None

Min X: 0.00

Max X: 120.00

Y-Axis Scale

☒ Linear Scale

☐ Square Root Scale

☐ Log Scale

Reset X-Axis

Accept Cancel

Figure 3. Plot Settings.

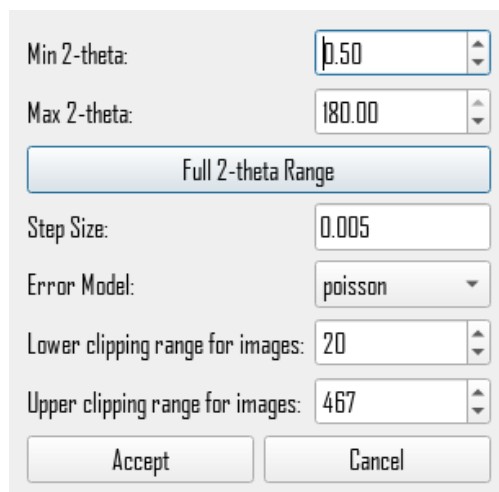
Use the plot toolbar (Figure 1, item 12) to zoom into a region, pan, go back to earlier views, or save the current plot as an image.

6. Importing and clearing data

- **File > Import Integrated Data** adds existing .xye files to the **Integrated Data** list. Several files can be selected at once.
- **File > Clear Data** removes all patterns and events from the session and clears the plot, after asking you to confirm. Your calibration file, SPEC file, image and output folders, scan numbers, and settings are kept, so you can integrate the next scan straight away. Files on disk are not touched. Live mode, if running, is stopped first.
- **File > Exit** closes the program.

7. Integration settings

Settings > Integration Settings controls how images are binned into a pattern. The defaults suit most measurements.



Min 2-theta: 0.50

Max 2-theta: 180.00

Full 2-theta Range

Step Size: 0.005

Error Model: poisson

Lower clipping range for images: 20

Upper clipping range for images: 467

Accept Cancel

Figure 4. Integration Settings.

Setting	Meaning
Min / Max 2-theta	The 2θ range written to the output file. Full 2-theta Range sets 0.5–180°; the file then contains only the range the detector actually covered.
Step Size	Width of the 2θ bins, in degrees (default 0.005).
Error Model	How the uncertainty column σ is calculated: poisson : $\sigma = \sqrt{I}$. Counting statistics only. azimuthal : σ = the spread (standard deviation) of the pixel values that fall in each 2θ bin. Spotty or textured rings, where pixels at the same 2θ differ a lot, get large errors; smooth, continuous rings get errors close to counting statistics.
Lower / Upper clipping range	Detector pixel columns outside this range (0–487) are left out, removing the detector edges. Defaults 20 and 467.

Important. Patterns integrated with the **azimuthal** error model in versions before October 2026 have a 2θ offset (about 1.6° for typical data) and a variance, not σ , in the third column. Re-integrate them with the current version. Data integrated with the **poisson** model are not affected.

8. Live mode

Live mode integrates each scan automatically as soon as it is complete and adds it to a live waterfall plot. Start it at the beginning of a run and leave it: no clicks are needed while you collect data.

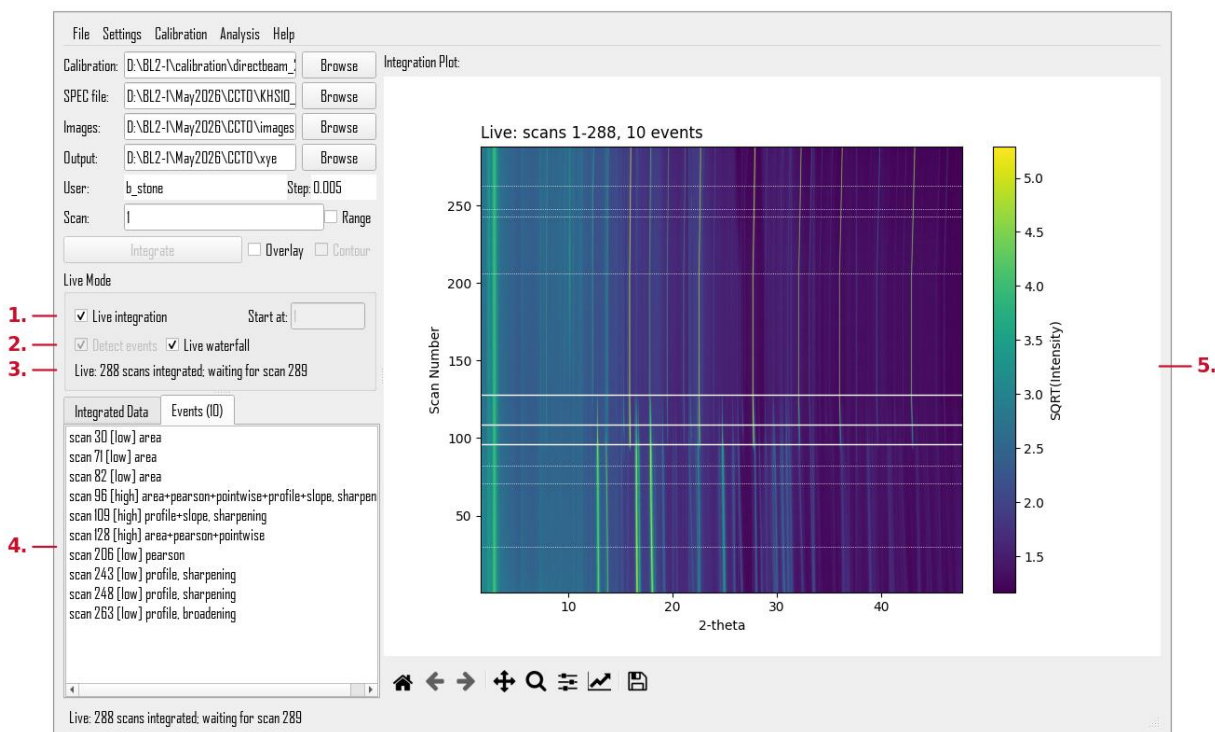


Figure 5. Live mode at the end of a 288-scan in-situ heating run, with detected events marked on the waterfall and listed in the Events tab.

1.	Live integration turns live mode on and off. Start at: leave blank to begin with the scan in progress (or the next scan); enter a scan number to also integrate earlier scans that are already collected.
2.	Detect events (Section 9) and Live waterfall (redraw the waterfall after each scan; untick to look at single patterns while live mode keeps running).
3.	Status: how many scans have been integrated and which scan is expected next.
4.	Events tab: events detected so far. Hover over an event for an explanation.
5.	Live waterfall: intensity vs 2θ and scan number. High-confidence events are solid lines, low-confidence events dotted.

8.1 Running live mode

1. Load the calibration file, SPEC file, image folder, and output folder (Section 4.1).
2. Leave **Start at** blank, or enter the first scan to include.
3. Tick **Live integration**. The status line shows which scan it is waiting for.

4. Collect data as usual. Each finished scan is integrated, saved as .xye, added to the data list, and drawn on the waterfall.
5. Untick **Live integration** to stop.

While live mode runs, the **Integrate** button is disabled.

8.2 When a scan is integrated

A scan is integrated once it is complete:

- its block in the SPEC file has all its points (for an ascan, the number of intervals plus one; for other scan types, when the next scan starts), and
- every image of the scan is present at full size, with its .pdi file.

A half-written image is never integrated, and no scan is integrated twice. Scans that never produce images, such as motor alignment scans, are skipped after 30 seconds so they do not hold up the queue. The SPEC file is checked every 2 seconds.

9. Detected events

An event is a scan where the diffraction pattern changed in a way that smooth thermal expansion does not explain. Detection does not know which phases are present: it flags **where** something changed, and you decide **what** changed. Events are suggestions, for example for where to start a new region in a sequential refinement; compare the patterns just before and after an event before relying on it.

9.1 Reading an event

scan 96 [high] area+pearson+pointwise+profile+slope, sharpening, 905 °C

Part	Meaning
scan 96	Where the change is centred. In live mode it is reported about 6 scans later, once the detectors can confirm it.
[high] / [low]	Confidence (below).
area+pearson+...	The detector families that responded (Section 9.3).
sharpening / broadening	Shown when peak widths changed.
905 °C	Temperature at the event, when the furnace temperature is recorded in the SPEC file.

9.2 Confidence

- **[high]**: two or more independent detector families agree, so the pattern changed in more than one way. Most likely a real transformation.
- **[low]**: only one family responded. It may be real (a subtle or gradual change that only one family sees) or a false alarm: sample movement, beam or intensity fluctuations, crowded peaks near the edge of the 2θ range, or noise early in a live run. Look at the patterns before deciding.

9.3 Detector families

Each pattern is compared with the pattern 3 scans earlier, allowing peaks to shift by up to $\pm 0.05^\circ 2\theta$ so that thermal expansion and contraction are not events.

Family	Measures	May indicate
area (area+ / area-)	Integrated intensity appearing or disappearing in a 2θ region; not affected by peak width changes.	area+: a phase forming or growing (reaction product, intermediate, new polymorph, crystallization). area-: a phase consumed, decomposing, melting, or becoming amorphous. Both: one phase converting into another. A loss and later a gain at the same positions: a reversible transition.
pointwise	The same comparison, point by point on vintensity.	Most sensitive to weak or minor phases. With profile but without area, usually a peak-shape change rather than a new phase.
profile	Peak sharpness at roughly	Sharpening: crystallite growth, annealing,

Family	Measures	May indicate
	constant area.	sintering, or the finest crystallites reacting first. Broadening: strain, defects, smaller crystallites, disorder, or a peak starting to split.
slope	How fast the whole pattern is moving away from the first scan.	Gradual changes: slow reactions, steadily changing phase fractions. Also changes in heating rate.
pearson	Abrupt change of the whole pattern between consecutive scans.	Fast events; also the start or end of a hold or of cooling. Misses changes in weak phases.

9.4 What is not detected

- A phase that is present but not changing. “No events during a hold” means nothing changed beyond the thresholds, not that a reaction is complete.
- Changes smaller than the noise, roughly a percent of the pattern's intensity.
- Smooth peak shifts (thermal expansion, gradual composition change) are ignored by design. A sudden lattice change larger than 0.05° can appear as an area or pointwise event.

9.5 Detecting events in data already loaded

Analysis > Detect Events in Selected Data runs the same detection on patterns already in the data list, whether integrated or imported. It uses the selected patterns, or all patterns if none are selected, and orders them by the scan number in their file names (..._scanN.xye). If the matching SPEC file is loaded, scan times and furnace temperatures (when recorded) are added to the events. At least 25 patterns are needed.

Because it can use the whole series as its reference, this usually gives fewer false alarms than live mode, which only has the scans collected so far. For finished runs, it is the better choice.

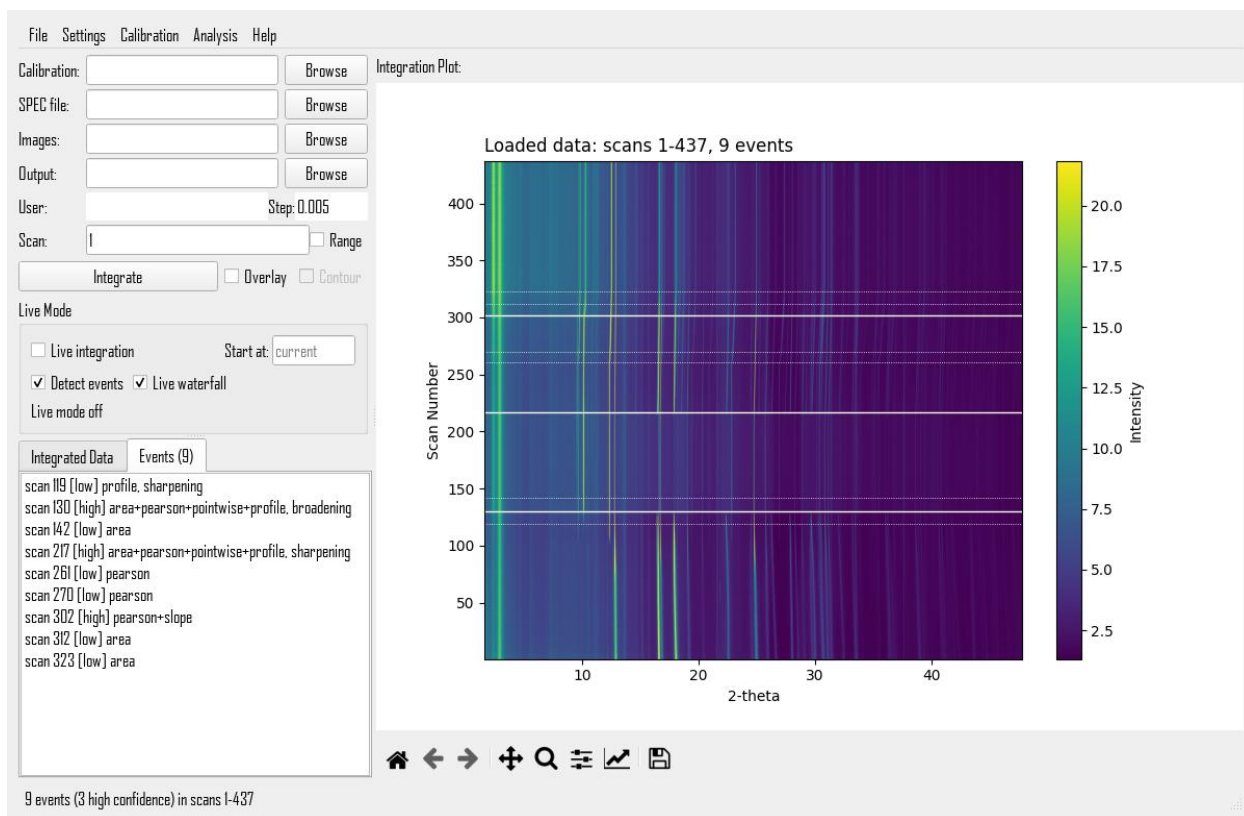


Figure 6. Events detected in a loaded 437-scan run. A reversible change appears as intensity lost at scan 130 (start of the hold) and regained at scan 217 (start of cooling).

Help > Understanding Detected Events gives this explanation inside the program, and hovering over any event in the Events tab explains that particular event.

10. Calibration

The calibration file gives the direct-beam position on the detector (pixel x, y) and the sample-to-detector distance, which together set the 2θ value of every pixel. It is made with the beamline calibration script from a direct-beam scan, and its contents look like:

```
direct_beam_x 264
direct_beam_y 79
Sample_Detector_distance_pixels 4118.07
Sample_Detector_distance_mm 708.308
```

Use the calibration measured for your detector setup; if the detector has been moved, a new calibration is needed. **Calibration > Run Calibration** is not available in the program yet; it reminds you to use the calibration script and load the .cal file it writes.

11. Troubleshooting

Problem	Likely cause	What to do
“No user name was found in the SPEC file”	The SPEC file has no User = line, or the wrong file was selected.	Select the SPEC file written for your measurements.
Integration fails: image not found	Image names must be <user>_<SPEC file name>_scan<N>_<kkkk>.raw (e.g. b_stone_KHS10_27D_CaCO3_CuO_anneal_scan1_0000.raw).	Check the Images folder and that the scan number exists.
Integration fails: unexpected RAW file size	An image is incomplete or not a Pilatus 100K .raw image.	Wait for collection to finish, or check the file.
Live mode is waiting and nothing happens	The scan in progress is not complete yet, its images are not all written, or Start at is beyond the last scan.	Check the status line; it names the scan it is waiting for.
Many low-confidence events early in a live run	Live detection has little history at the start.	Afterwards, run Analysis > Detect Events in Selected Data on the whole run.
Peaks in the 2 θ axis are off	Wrong calibration file for this detector setup.	Use the calibration measured for these scans.
Azimuthal-mode data from older versions look shifted	Bug in versions before October 2026.	Re-integrate with the current version (Section 7).

12. Menu reference

Menu	Items
File	Import Integrated Data · Clear Data · Exit
Settings	Plot Settings (Section 5) · Integration Settings (Section 7)
Calibration	Run Calibration (not yet available; see Section 10)
Analysis	Detect Events in Selected Data (Section 9.5)
Help	Live Mode Guide · Understanding Detected Events · Open Manual PDF · About

Help > Open Manual PDF opens manual.pdf from the folder the program is in. If it is not there, it opens the copy on GitHub instead, which needs access to the repository.